Deliverable 5: 5-Minute Demonstration Video Note

Project: Never Lose a Heartbeat Data Pipeline

Candidate: Sehaj (GitHub: `sehaj141`)

Repository: https://github.com/sehaj141/never-lose-a-heartbeat

Video Duration: Under 5 minutes (MP4)

Video Contents & Timestamps

The video demonstration was recorded following the official evaluation runbook:

- **0:00 – 1:00 (Baseline Online Mode):**
- Demonstrates concurrent execution of `biosignal-mock`, `pipeline_pkg/cloud_server.py`, and `pipeline start`.
- Verifies live streaming status using `pipeline status` showing `"mode": "online"` with real-time cloud data ingestion.

- **1:00 – 2:00 (Simulated Network Outage):**
- Network connection to cloud server is severed.
- Demonstrates the pipeline watchdog detecting connection failure and transitioning to `"mode": "offline"` within 10 seconds.
- Demonstrates local persistence and growing backlog counter.

- **2:00 – 3:15 (Network Restoration & Catch-Up):**
- Cloud server connectivity restored.
- Demonstrates automatic transition back to `"mode": "online"` within 10 seconds.
- Shows high-throughput backlog replay draining to 0 without dropped samples or duplicates.

- **3:15 – 4:00 (Process Kill Resilience):**
- Abruptly terminates the pipeline daemon with `kill -9` (SIGKILL).
- Restarts daemon and executes `pipeline status` to verify SQLite WAL crash recovery with zero corrupted records.

- **4:00 – 5:00 (Power Severance & Ground Truth Audit):**
- Hard kernel reset simulation (`sysrq-trigger`).

- Executes `pipeline export --from device` and verifies Section 3 window processing results via `pipeline results`.